Scalable Bayesian Synthesis of Relational Enterprise Databases: Algorithms and Complexity

Data Synthesisation project

5 October 2026

Formalisation of the layer-1 method list (`research/layer1-final-audit.md`, §5)

Abstract

We formalise the method list selected in layer 1 of the Data Synthesisation project as seven algorithms with stated models, update equations, and time and space complexity in big-O notation. The pipeline (i) summarises each table in a single parallel pass with mergeable sketches and a stratified sample, (ii) fits a two-stage Bayesian Gaussian copula whose marginals are pinned by sketch quantiles and whose correlation is learned from the sample, (iii) fits an infinite-relational-model-style latent-segment model across all dimensions of a star or snowflake schema using only the sparse sufficient statistics of one `GROUP BY` per fact table, (iv) merges surplus segments with exact closed-form marginal likelihoods, (v) generates a referentially intact synthetic database of user-chosen scale top-down, (vi) controls disclosure through a person-level weighted pseudo-posterior that becomes ε-differentially private under a stated bound, and (vii) releases high-dimensional sensitive blocks through random orthonormal projection with the Gaussian mechanism. The per-iteration cost of model fitting is $O(\sum_r \mathrm{nnz}_r + \sum_e n_e K_e L_e)$, independent of the number of fact rows. Every proposition that a test can check was verified numerically (14 checks, all passing); the validation also found and corrected seven errors or gaps in the layer-1 documents (Section 10).

Contents

1 Scope and setting **2**

2 Algorithm 1: prior discovery in one pass **2**

3 Algorithm 2: two-stage Bayesian Gaussian copula **3**

4 Algorithm 3: relational latent-segment model **5**

5 Algorithm 4: merging surplus segments **6**

6 Algorithm 5: top-down synthesis at user-chosen scale **7**

7 Algorithm 6: person-level privacy **8**

8 Algorithm 7: random orthonormal projection for sensitive high-dimensional blocks **9**

9 Outer loop, evaluation, and end-to-end cost **10**

10 Validation **10**

1 Scope and setting

Mapping to the layer-1 list. Table 1 maps each algorithm to the item it formalises in the authoritative layer-1 list. Items 8–10 (tuning, baselines, evaluation) are an outer loop around the method rather than parts of it; Section 9 states their cost only.

Algorithm	Formalises (layer-1 list, §5)	Status in layer 1
1 Prior discovery	#1 sketches, stratified sample, sequential updating	core
2 Two-stage copula	#2 per-table engine; #4 within-segment model	core
3 Relational latent segments	#3 relational engine; #5 temporal facts (extension)	core
4 Sufficient-statistic merge	#3 "split–merge" requirement	core
5 Top-down synthesis	#3 generation, user-controlled scale	core
6 Person-level pseudo-posterior	#6 privacy dial and DP option	core / option
7 Random orthonormal projection	#7 RP for sensitive high-dimensional blocks	option

Table 1: Algorithms and the layer-1 items they formalise.

Schema. A database is a directed acyclic graph $G = (\mathcal{T}, \mathcal{F})$ of tables, with an edge $T \to T'$ when T' holds a foreign key to T. Tables split into *entity types* $e \in \mathcal{E}$ (dimension tables, including snowflake levels) with entity sets U_e, $n_e = |U_e|$, and *relations* $r \in \mathcal{R}$ (fact and bridge tables) of arity a_r whose rows reference a tuple $\mathbf{u} = (u_1, \dots, u_{a_r}) \in U_{e_1} \times \cdots \times U_{e_{a_r}}$. Relation r has N_r rows and d_r numeric measures; entity type e has p_e attributes. A snowflake level is an entity type $h(e)$ referenced by e; a self-reference is an edge $e \to e$ (for example employee $\to$ manager).

Sufficient statistics. For a relation r the *tuple statistics* are, for every tuple $\mathbf{u}$ with at least one row,

$$c_r(\mathbf{u}) = \#\{\text{rows with key } \mathbf{u}\}, \qquad s_r(\mathbf{u}) = \sum y, \qquad q_r(\mathbf{u}) = \sum y^2 \;\; (\text{or } \sum \mathbf{y}\mathbf{y}^\top \text{ for } d_r > 1),$$

which one `GROUP BY` over the key columns returns. Their number is $\mathrm{nnz}_r \le N_r$. Tuples with $c_r(\mathbf{u}) = 0$ are never stored.

Cost model. Complexities are in the word-RAM model. We separate (a) *one-off* costs paid inside the source database (scans and aggregates, parallel over P workers), (b) *per-iteration* costs of the sampler, which never touches raw fact rows, and (c) *generation* costs, linear in the output. K_e denotes the (over-specified) number of segments of type e, and I the number of MCMC iterations.

2 Algorithm 1: prior discovery in one pass

For every column j of a table with N rows and p columns we compute in one parallel scan:

- continuous column: a KLL quantile sketch [14] with parameter k, whose normalised rank error is at most $\epsilon_k = O(1/k)$ with high probability, and which is *mergeable*;
- discrete column with V_j levels: exact level counts (a hash aggregate of size $O(V_j)$); for very large V_j, a Count-Min sketch [4] of width $\lceil e/\epsilon \rceil$ and depth $\lceil \ln(1/\delta) \rceil$, which overestimates any count by at most ϵN with probability $1 - \delta$;
- key columns: a HyperLogLog sketch [7] with m registers (relative standard error $\approx 1.04/\sqrt{m}$) for distinct counts;
- a stratified reservoir sample of n rows; strata are user-chosen (for example a segment proxy or a time bucket).

Algorithm 1 Prior discovery (per table)

Input: table partitions $D_1, \dots, D_P$; sketch parameters k, ϵ, δ, m; sample size n with proportional allocation over strata

Output: per-column sketches $\{\mathcal{K}_j\}$, sample $\mathcal{S}$, conjugate posteriors $\{p(\theta \mid D)\}$

for each worker $w = 1..P$ **in parallel do**
  for each row $x \in D_w$ **do**
    update $\mathcal{K}_j^{(w)}$ with x_j for $j = 1..p$ ▷ KLL / exact counts / Count-Min / HLL
    offer x to the reservoir of its stratum ▷ Algorithm R or L
    accumulate exponential-family sufficient statistics $t^{(w)} \mathrel{+}= T(x)$
$\mathcal{K}_j \leftarrow \textsc{Merge}(\mathcal{K}_j^{(1)}, \dots, \mathcal{K}_j^{(P)})$; merge reservoirs; $t \leftarrow \sum_w t^{(w)}$
$p(\theta \mid D) \leftarrow$ conjugate update of prior $p_0(\theta)$ with t ▷ sequential: prior ← last posterior

Sequential updating. For conjugate exponential families $p(x \mid \theta) = h(x) \exp\{\theta^\top T(x) - A(\theta)\}$ with prior $p_0(\theta) \propto \exp\{\theta^\top \chi_0 - \nu_0 A(\theta)\}$, the posterior after batches $D_1, \dots, D_b$ has $\chi_b = \chi_0 + \sum_{i \le b} \sum_{x \in D_i} T(x)$ and $\nu_b = \nu_0 + \sum_{i \le b} |D_i|$. Using the batch-$b$ posterior as the batch-$(b{+}1)$ prior therefore gives exactly the full-data posterior, in any order and at any partitioning. This is the "posterior → prior" loop of the original request, made exact.

Complexity. KLL updates cost $O(\log k)$ amortised (each compaction sorts its buffer and an item reaches level h with probability 2^{-h}), Count-Min $O(\ln(1/\delta))$, HLL $O(1)$, Algorithm R $O(1)$. Let $c_{\text{sk}} = \log k + \ln(1/\delta)$.

$$\text{Time } O\Big(\tfrac{Np\,c_{\text{sk}}}{P} + P\,p\,(k + \epsilon^{-1} \ln \delta^{-1} + m)\Big), \qquad \text{Space } O\Big(p\,(k + \epsilon^{-1} \ln \delta^{-1} + m) + \textstyle\sum_j V_j + np\Big).$$

The second time term is the merge. Space is independent of N except through the exact level tables.

Remark 1 (Random projection is not used here). Layer 1 (experiment 1) found stratified sampling 20–700× faster than random projection at equal accuracy for estimating priors, so random projection is excluded from this step.

3 Algorithm 2: two-stage Bayesian Gaussian copula

Model. For a table with mixed-type columns $j = 1..p$ and marginal CDFs F_j,

$$\mathbf{z}_i \overset{\text{iid}}{\sim} \mathcal{N}_p(\mathbf{0}, C), \qquad x_{ij} = F_j^{-1}(\Phi(z_{ij})), \qquad C \sim \text{LKJ}(\eta),$$

with $F_j^{-1}(u) = \inf\{x : F_j(x) \ge u\}$ and C a correlation matrix. Because F_j is a step function for discrete columns, x_{ij} only tells us that z_{ij} lies in an interval.

Stage 1: anchors from the full data. From Algorithm 1 each column receives latent *anchors* $-\infty = a_{j0} < a_{j1} < \cdots < a_{jm_j} = \infty$:

- discrete column with sorted levels $v_1 < \cdots < v_{V_j}$: $a_{jq} = \Phi^{-1}(\hat{F}_j(v_q))$ from *exact* level counts, $m_j = V_j$; bin q is the level v_q;
- continuous column: $a_{jq} = \Phi^{-1}(\tau_q)$ for a probability grid $\tau_1 < \cdots < \tau_{m_j - 1}$, and bin q is $(\hat{Q}_j(\tau_{q-1}), \hat{Q}_j(\tau_q)]$ where $\hat{Q}_j$ is the KLL quantile function.

Observation x_{ij} in bin q implies $z_{ij} \in (a_{j,q-1}, a_{jq}] =: (\ell_{ij}, u_{ij}]$. This is the extended quantile likelihood of Feldman and Kowal [6], which coarsens each column to bins whose latent boundaries are fixed by externally known quantiles; with exact level counts it is lossless for discrete columns.

Algorithm 2 Two-stage Bayesian Gaussian copula (one table or one segment)

Input: anchors $\{a_{jq}\}$ (Stage 1); sample $\mathcal{S}$; LKJ shape η; iterations I; Metropolis steps T, step size h

Output: posterior draws $C^{(1..I)}$; synthetic rows on request

$(\ell_{ij}, u_{ij}] \leftarrow$ bin of x_{ij}; initialise z_{ij} inside its bin; $y \leftarrow 0$

for $t = 1..I$ **do**

  $C \leftarrow L(y)L(y)^\top$; $\Omega \leftarrow C^{-1}$ ▷ $O(p^3)$

  for $j = 1..p$ **do** ▷ Step 1, $O(np)$ per column

    $\mu_{\cdot j} \leftarrow -(Z\,\Omega_{\cdot j} - Z_{\cdot j}\Omega_{jj})/\Omega_{jj}$; $\sigma_j \leftarrow \Omega_{jj}^{-1/2}$

    $z_{ij} \sim \mathrm{TN}(\mu_{ij}, \sigma_j^2;\ \ell_{ij}, u_{ij})$ for all i ▷ rows sharing a bin share bounds

  $S \leftarrow Z^\top Z$ ▷ $O(np^2)$

  for $\tau = 1..T$ **do** ▷ Step 2, $O(p^3)$ per proposal, independent of n

    $y' \leftarrow y + h\,\xi$, $\xi \sim \mathcal{N}(0, I)$; accept with prob. $\min\{1, \exp[\pi(y') - \pi(y)]\}$ using (1), the Beta prior and Jacobian

Synthesis of M rows: draw C from the chain; $\mathbf{z}^* \sim \mathcal{N}_p(0, C)$; $x_j^* = \hat{Q}_j(\Phi(z_j^*))$ (KLL) or the level whose bin contains z_j^*

Stage 2: correlation from the sample. The likelihood of the stratified sample $\mathcal{S}$ (n rows) is

$$p(\mathcal{S} \mid C) = \int_{\prod_{ij}(\ell_{ij}, u_{ij}]} \prod_{i=1}^{n} \phi_p(\mathbf{z}_i; \mathbf{0}, C)\, d\mathbf{z}.$$

We sample it by data augmentation (Algorithm 2). Given Z, the posterior of C depends on Z only through $S = Z^\top Z$:

$$\log p(C \mid Z) = \text{const} + (\eta - 1)\log|C| - \tfrac{n}{2}\log|C| - \tfrac{1}{2}\operatorname{tr}(C^{-1}S). \tag{1}$$

We parameterise $C = LL^\top$ through the $p(p-1)/2$ canonical partial correlations w of a C-vine [16]: $L_{11} = 1$, $L_{i1} = w_{i1}$, $L_{ij} = w_{ij}\sqrt{1 - \sum_{k<j} L_{ik}^2}$ for $1 < j < i$, and $L_{ii} = \sqrt{1 - \sum_{k<i} L_{ik}^2}$. Under LKJ($\eta$) the w_{ij} are independent, and $(w_{ij} + 1)/2 \sim \mathrm{Beta}(b_j, b_j)$ with $b_j = \eta + (p - 1 - j)/2$ for conditioning level $j = 1, \ldots, p-1$ (index j of the column of L). We run random-walk Metropolis on $y = \operatorname{atanh}(w)$, including the Jacobian $\prod(1 - w^2)$.

Complexity.

$$\text{Fit: time } O(I\,(np^2 + Tp^3)),\ \text{space } O(np + p^2); \qquad \text{Synthesis: } O(p^3 + M(p^2 + p\log k)).$$

Step 1 draws np truncated normals, each needing an $O(p)$ conditional mean, hence $O(np^2)$ per sweep. A rank-r factor model for C, as used by Feldman and Kowal, reduces this to $O(npr + pr^2)$.

Remark 2 (Why two stages). Hoff's extended rank likelihood [9] handles ties correctly but must update all Np latent values every iteration; on a full fact table that is $\Theta(Np^2)$ per sweep. Pinning the marginals with full-data sketches lets the dependence be learned from $n \ll N$ rows. Layer 1 (experiment 1) measured 0.8% relative error at $n = 2 \times 10^5$.

Remark 3 (Anchors for discrete columns). Anchors must be strictly increasing, or a bin is empty and the likelihood is zero. KLL ranks of two rare adjacent levels can coincide within the sketch error, which our first test run hit (Section 10). Discrete columns therefore use exact level counts, which cost $O(V_j)$ space; KLL is used only for continuous columns, on a grid with gaps larger than $2\epsilon_k$.

Remark 4 (Sample weighting). Equation (1) assumes the sample is self-weighting (proportional allocation). With unequal inclusion probabilities π_i, replace S by $\sum_i w_i \mathbf{z}_i \mathbf{z}_i^\top$ and n by $\sum_i w_i$, with $w_i \propto 1/\pi_i$ normalised to sum to n (survey pseudo-posterior [19]).

4 Algorithm 3: relational latent-segment model

Generative model. We state it for a binary relation r between types e_1 (index u, segments $k \le K$) and e_2 (index v, segments $l \le L$); Remark 5 gives the general case.

$$\begin{aligned}
\pi &\sim \mathrm{Dir}(\gamma/K \cdot \mathbf{1}_K), & z_u &\sim \mathrm{Cat}(\pi), & x_u \mid z_u = k &\sim f_{e_1}(\cdot \mid \theta_k),\\
w &\sim \mathrm{Dir}(\gamma/L \cdot \mathbf{1}_L), & z_v &\sim \mathrm{Cat}(w) \text{ or } \mathrm{Cat}(\psi_{t_{h(v)}}), & x_v \mid z_v = l &\sim f_{e_2}(\cdot \mid \theta'_l),\\
\eta_{kl} &\sim \mathrm{Gamma}(a_\eta, b_\eta), & c(u,v) &\sim \mathrm{Pois}(\eta_{z_u z_v}), & \sigma^{-2} &\sim \mathrm{Gamma}(a_\sigma, b_\sigma),\\
m_{kl} &\sim \mathcal{N}(m_0, s_0^2), & y &\sim \mathcal{N}(m_{z_u z_v}, \sigma^2),
\end{aligned}$$

with $c(u,v)$ independent over all $(u,v) \in U_{e_1} \times U_{e_2}$, one measure y per row, and $\mathrm{Cat}(\psi_{t_{h(v)}})$ used when v has a snowflake parent $h(v)$ with segment t. This is a Poisson stochastic block model, the count analogue of the infinite relational model [15, 18], with K, L over-specified and sparse Dirichlet priors so that surplus segments can empty [17]. f_e is the copula of Algorithm 2 per segment, a Gaussian, or a Dirichlet–categorical model.

Block statistics. With one-hot matrices $Z_1 \in \{0,1\}^{n_1 \times K}$, $Z_2 \in \{0,1\}^{n_2 \times L}$ and the sparse tuple-statistic matrices $M = [c(u,v)]$, $S = [s(u,v)]$, $Q = [q(u,v)]$:

$$C^{\mathrm{row}} = M Z_2 \in \mathbb{R}^{n_1 \times L}, \quad S^{\mathrm{row}} = S Z_2,$$

$$C^{\mathrm{blk}} = Z_1^\top C^{\mathrm{row}},\ S^{\mathrm{blk}},\ Q^{\mathrm{blk}} \in \mathbb{R}^{K \times L}, \quad n_k = \mathbf{1}^\top Z_1,\ m_l = \mathbf{1}^\top Z_2.$$

Proposition 1 (Segment conditional from sufficient statistics)**.** *Given all other variables, for $u \in U_{e_1}$,*

$$\begin{aligned}
\log p(z_u = k \mid \cdot) = \mathit{const}_u &+ \log \pi_k + \log f_{e_1}(x_u \mid \theta_k)\\
&+ \sum_{l=1}^{L} \Big[C^{\mathrm{row}}_{ul} \log \eta_{kl} - \eta_{kl} m_l \Big] + \frac{1}{\sigma^2} \sum_{l=1}^{L} \Big[S^{\mathrm{row}}_{ul} m_{kl} - \tfrac{1}{2} C^{\mathrm{row}}_{ul} m_{kl}^2 \Big].
\end{aligned} \tag{2}$$

The z_u, $u \in U_{e_1}$, are conditionally independent given $(z_2, \text{parameters})$, and so can be drawn in parallel.

Proof. The terms of the joint density that involve z_u are $\pi_{z_u} f(x_u \mid \theta_{z_u})$, the Poisson factors for all $v \in U_{e_2}$, *including those with* $c(u,v) = 0$, and the Normal factors of u's rows. Summing the Poisson log-pmf over v and grouping by $z_v = l$ gives $\sum_l [(\sum_{v: z_v = l} c(u,v)) \log \eta_{z_u l} - m_l \eta_{z_u l}] - \sum_v \log c(u,v)!$, and the last sum does not depend on z_u. For the Normal rows, $\sum -\frac{(y-m)^2}{2\sigma^2} = -\frac{1}{2\sigma^2}(Q - 2mS + Cm^2)$ grouped by l, and the Q term does not depend on z_u. Conditional independence holds because no factor couples z_u and $z_{u'}$ once z_2 and the parameters are fixed. □

Zero cells enter only through the exposure m_l. This is why the sampler never enumerates the $n_1 n_2$ pairs.

Complexity. The statistics build is one aggregate: $O(N_r)$ expected time with hash aggregation, or $O(N_r \log N_r)$ sort-based, producing nnz_r tuples. Per iteration,

$$\text{time } O\big(\mathrm{nnz}_r + (n_1 + n_2)KL + n_1 p_1^2 + n_2 p_2^2\big), \qquad \text{space } O\big(\mathrm{nnz}_r + n_1 K + n_2 L\big).$$

This is independent of N_r (repeat purchases collapse into one tuple) and of the $n_1 n_2$ possible pairs. Parallelising the two z-updates over P workers divides the dominant terms by P.

Algorithm 3 Blocked Gibbs for the relational latent-segment model (binary relation)

Input: tuple statistics M, S, Q (one `GROUP BY`); attributes x; caps K, L; priors; I
Output: posterior draws of $(z_1, z_2, \pi, w, \eta, m, \sigma^2, \theta)$

warm start: $z_2 \leftarrow k$-means++ on (attributes, $\log(1+\deg)$, mean measure) of e_2; then z_1 likewise using MZ_2 ▷ $O(nKd\,I_{\mathrm{km}})$

for $t = 1..I$ **do**

  $C^{\mathrm{blk}}, S^{\mathrm{blk}}, Q^{\mathrm{blk}}, n_k, m_l \leftarrow$ scatter-add over the non-zeros ▷ $O(\mathrm{nnz} + n_1 + n_2)$

  $\eta_{kl} \sim \mathrm{Gamma}(a_\eta + C^{\mathrm{blk}}_{kl},\ b_\eta + n_k m_l)$

  $m_{kl} \sim \mathcal{N}(\lambda_{kl}^{-1}(m_0/s_0^2 + S^{\mathrm{blk}}_{kl}/\sigma^2),\ \lambda_{kl}^{-1}),\ \lambda_{kl} = 1/s_0^2 + C^{\mathrm{blk}}_{kl}/\sigma^2$

  $\sigma^{-2} \sim \mathrm{Gamma}\left(a_\sigma + \frac{1}{2}\sum_{kl} C^{\mathrm{blk}}_{kl},\ b_\sigma + \frac{1}{2}\sum_{kl}(Q^{\mathrm{blk}}_{kl} - 2m_{kl}S^{\mathrm{blk}}_{kl} + C^{\mathrm{blk}}_{kl}m_{kl}^2)\right)$

  $\pi \sim \mathrm{Dir}(\gamma/K + n_{1..K})$; $w \sim \mathrm{Dir}(\gamma/L + m_{1..L})$; $\theta_k \mid \cdot$ conjugate per segment ▷ $O(n_e p_e^2)$

  z_u for all $u \in U_{e_1}$ **in parallel** from (2) ▷ $O(\mathrm{nnz} + n_1 KL)$

  z_v for all $v \in U_{e_2}$ **in parallel**, symmetric form with $M^\top$, plus $\log \psi_{t,l}$ for its snowflake parent

  snowflake parents b: $\log p(t_b = t \mid \cdot) = \log \rho_t + \sum_l B_{bl} \log \psi_{tl}$, $B_{bl} = \#\{v : h(v) = b, z_v = l\}$; $\psi_t \sim \mathrm{Dir}(\cdot)$

Remark 5 (General schemas). For a relation of arity a with partner types $e_1..e_a$, the row statistic of $u \in U_{e_1}$ is a tensor in $\mathbb{R}^{K_2 \times \cdots \times K_a}$ and (2) sums over all partner-segment tuples, with exposure $\prod_{i \geq 2} m^{(i)}_{l_i}$. The cost is $O(a \cdot \mathrm{nnz}_r + \sum_i n_{e_i} \prod_{i'} K_{e'_i})$. An entity type that appears in several relations adds one such sum per relation to its conditional, so a full iteration costs $O(\sum_r a_r\, \mathrm{nnz}_r + \sum_e n_e \sum_{r \ni e} \prod_{e' \in r} K_{e'})$. Snowflake chains recurse: each level's segment is the prior of the level below, as in the last line of Algorithm 3.

Remark 6 (Self-referencing trees; design only). For an edge $e \to e$ (a tree with parent map par), let $z_u \mid z_{\mathrm{par}(u)} \sim \mathrm{Cat}(\psi_{z_{\mathrm{par}(u)}})$ and the number of children $N_u \sim \mathrm{Pois}(\lambda_{z_u})$, a multi-type Galton–Watson process. Then $\log p(z_u = k \mid \cdot) = \log \psi_{z_{\mathrm{par}(u)},k} + \sum_{c:\mathrm{par}(c)=u} \log \psi_{k,z_c} + N_u \log \lambda_k - \lambda_k + \log f(x_u \mid \theta_k)$. Nodes at even depth are conditionally independent given nodes at odd depth, so a two-colour blocked Gibbs sweep costs $O(n_e K)$. This has not been tested yet.

Remark 7 (Temporal facts; design only). With time buckets $b = 1..B$ of length Δ_b, let $c(u, v, b) \sim \mathrm{Pois}(\eta_{kl}\lambda_{kb}\Delta_b)$ with $\lambda_{kb} \sim \mathrm{Gamma}$ under the identifiability constraint $\sum_b \lambda_{kb}\Delta_b = \sum_b \Delta_b$, for example via a random-walk or seasonal prior on $\log \lambda_{k\cdot}$. The statistics come from `GROUP BY` (u, v, b), with $\mathrm{nnz}_{r,T} \geq \mathrm{nnz}_r$ tuples, and an iteration costs $O(\mathrm{nnz}_{r,T} + (n_1 + n_2)KLB)$. SDV PAR [22] is the baseline.

Remark 8 (Mixing). Layer 1 (experiment 3) found that an exact-K sampler with random initialisation can stall in a mode where two true segments merge and another splits. Algorithm 3 therefore uses over-specified caps, sparse Dirichlet priors, data-driven warm starts and several chains, followed by Algorithm 4.

5 Algorithm 4: merging surplus segments

Over-specified fits leave duplicate segments. These are harmless for synthesis, but they hurt interpretability. All block statistics are additive under a merge, and the blocks have closed-form marginal likelihoods, so each candidate merge can be scored exactly in $O(L + p)$ time.

Proposition 2 (Block marginal likelihoods)**.** *For a block with total count C over exposure $E = n_k m_l$ pairs and prior $\eta \sim \mathrm{Gamma}(a, b)$,*

$$\log \int \prod_{pairs} \mathrm{Pois}(c \mid \eta)\ \mathrm{Gamma}(\eta \mid a, b)\, d\eta = a \log b - \log \Gamma(a) + \log \Gamma(a + C) - (a + C)\log(b + E) - \sum \log c!. \tag{3}$$

Algorithm 4 Greedy sufficient-statistic merge

Input: final (or MAP) z_1, z_2; block statistics; α; plug-in $\hat{\sigma}^2$
Output: merged partitions

repeat
  for $e \in \{e_1, e_2\}$ **do** ▷ partner partition fixed
    drop empty segments; compute $\Delta_{kk'}$ for all pairs ▷ $O(K^2(L+p))$
    while $\max \Delta_{kk'} > 0$ **do**
      merge the arg-max pair: add rows/columns of the block statistics; update affected Δ ▷ $O(K(L+p))$
until no merge in a full pass

For C measures with sum S, sum of squares Q, known σ^2 and $m \sim \mathcal{N}(m_0, \sigma^2/\kappa_0)$ (that is, $\kappa_0 = \sigma^2/s_0^2$),

$$\log p(y_{1..C}) = -\tfrac{C}{2}\log(2\pi\sigma^2) + \tfrac{1}{2}\log\tfrac{\kappa_0}{\kappa_0+C} - \tfrac{1}{2\sigma^2}\Big[Q + \kappa_0 m_0^2 - \tfrac{(S+\kappa_0 m_0)^2}{\kappa_0+C}\Big]. \tag{4}$$

Proof. (3): the integrand is $\frac{b^a}{\Gamma(a)}\eta^{a+C-1}e^{-(b+E)\eta}/\prod c!$, a Gamma kernel. (4): $\mathbf{y} \sim \mathcal{N}(m_0\mathbf{1}, \sigma^2(I + \mathbf{1}\mathbf{1}^\top/\kappa_0))$. Apply the matrix determinant lemma and Sherman–Morrison and collect terms. □

The score of a partition z_e of type e, with the partner partition held fixed, is

$$\mathcal{J}(z_e) = \underbrace{K\log\alpha + \textstyle\sum_k \log\Gamma(n_k)}_{\text{CRP prior, up to const.}} + \sum_{k,l}\big[(3)(C_{kl}^{\text{blk}}, n_k m_l) + (4)(C_{kl}^{\text{blk}}, S_{kl}^{\text{blk}}, Q_{kl}^{\text{blk}})\big] + \sum_k \log p(x_{U_k}),$$

where the last term is the attribute marginal likelihood. Merging k and k' changes $\mathcal{J}$ by

$$\Delta_{kk'} = \log\tfrac{\Gamma(n_k+n_{k'})}{\alpha\,\Gamma(n_k)\Gamma(n_{k'})} + \sum_l \big[g(C_{kl}+C_{k'l}, (n_k+n_{k'})m_l) - g(C_{kl}, n_k m_l) - g(C_{k'l}, n_{k'} m_l)\big] + \ldots,$$

with g the per-block terms; every argument is a sum of existing statistics.

Complexity. $O(K^2(L+p))$ to initialise and $O(K(L+p))$ per merge, so $O(K^2(L+p))$ per type and pass. Each accepted merge strictly increases $\mathcal{J}$ and the number of segments is finite, so the procedure terminates. It needs no access to the data. The greedy rule gives a MAP-style point estimate. For a posterior over K, the same Δ is the target ratio inside Jain–Neal split–merge Metropolis–Hastings moves [12].

6 Algorithm 5: top-down synthesis at user-chosen scale

Proposition 3 (Poisson splitting)**.** *Let $N_{kl} \sim \text{Pois}(|A||B|\eta)$ and allocate each of the N_{kl} events to a pair drawn uniformly with replacement from $A \times B$. The resulting pair counts are independent* $\text{Pois}(\eta)$ *variables.*

Proof. Conditionally on N_{kl} the counts are multinomial with equal cell probabilities. A Poisson total with multinomial thinning gives independent Poisson cells with means $|A||B|\eta \cdot \frac{1}{|A||B|}$ (the colouring theorem). □

Proposition 4 (Scale rule)**.** *If the partner type e_2 is resized from m_l to m_l^* entities per segment, setting $\eta_{kl}^* = \eta_{kl}\, m_l/m_l^*$ keeps the distribution of each e_1 entity's row count, $\sum_v c(u,v) \mid z_u = k \sim \text{Pois}(\sum_l \eta_{kl} m_l)$, unchanged. Without the rule, mean rows per e_1 entity scale by m_l^*/m_l.*

Proof. $\sum_l \eta_{kl}^* m_l^* = \sum_l \eta_{kl} m_l$, and sums of independent Poissons are Poisson. □

The user names the *anchor* side whose per-entity cardinality must be preserved (usually the "one" side, such as customers). The table then grows linearly with that side's scale: $\mathbb{E}N_r^* = (n_{e_1}^*/n_{e_1})\,\mathbb{E}N_r$ when segment proportions are kept.

Algorithm 5 Top-down synthesis

Input: one posterior draw Θ (or D draws for multiple synthetic copies); target sizes n_e^*; anchor side per relation

Output: synthetic database with the source schema and new surrogate keys

1: **for** entity types e in topological order of G **do**
2: keys $1..n_e^*$; $z_u^* \sim \mathrm{Cat}(\pi_e)$, or $\mathrm{Cat}(\psi_{t^*})$ given the synthetic parent's segment
3: $x_u^* \sim f_e(\cdot \mid \theta_{z_u^*})$ ▷ Algorithm 2, per segment
4: **for** each relation r, each block (k, l) **in parallel do**
5: $\eta_{kl}^* \leftarrow \eta_{kl}\, m_l / m_l^*$ (anchor e_1); $N_{kl}^* \sim \mathrm{Pois}(n_k^* m_l^* \eta_{kl}^*)$
6: draw N_{kl}^* pairs uniformly from $U_k^* \times U_l^*$; measures $y \sim \mathcal{N}(m_{kl}, \sigma^2)$ (or the segment copula)
7: stream rows to the target schema by bulk load

Complexity. Time $O(\sum_e n_e^*(K_e + c_{f_e}) + \sum_r (K_r L_r + a_r \mathbb{E}N_r^*))$, where c_{f_e} is the per-row cost of the attribute model ($O(p^2 + p \log k)$ for the copula). Space is $O(\sum_e n_e^*)$ for keys and segment labels; facts are streamed. Every foreign key is drawn from an existing synthetic key, so referential integrity holds by construction. Business rules from the relationship file enter as structural zeros (blocks with $\eta_{kl} \equiv 0$ or excluded levels) and as deterministic post-hoc derivations.

7 Algorithm 6: person-level privacy

Unit. A *person* i owns the set D_i of all rows that link to them across all tables. Neighbouring databases differ by adding or removing one person's D_i. Row-level guarantees undercount the risk to a customer with thousands of fact rows.

Weighted pseudo-posterior. Given weights $\alpha_i \in [0, 1]$,

$$p_\alpha(\Theta \mid D) \propto p(\Theta) \prod_i \exp\{\alpha_i\, \tilde{\ell}_i(\Theta)\}, \qquad \tilde{\ell}_i = \ell_i - c_i, \quad \ell_i(\Theta) = \log p(D_i \mid \Theta),$$

where c_i is any constant depending only on D_i (it cancels in the normaliser). In a latent-segment model ℓ_i is the person's marginal likelihood with their own segment summed out, $\log \sum_k \pi_k f(x_i \mid \theta_k) \prod_l \{\cdots\}$ from (2). The sufficient-statistic structure survives: weights multiply each person's tuple statistics, so Algorithm 3 runs unchanged on weighted statistics.

- **Risk dial (default, no formal guarantee).** α_i decreases in a measured disclosure risk $r_i \in [0, 1]$, following the risk-weighted pseudo-posterior of Hu, Savitsky and Williams [10].
- **Formal DP (option).** Restrict Θ to a compact box Θ_B (bounded η, m, σ), bound contributions (at most $r_{\max}$ rows per person per table and values clipped to a box, as in [21]), take $c_i = \frac{1}{2}(\sup_{\Theta_B} \ell_i + \inf_{\Theta_B} \ell_i)$, $\mathrm{osc}_i = \frac{1}{2}(\sup - \inf)$, and set $\alpha_i = \min(1, B/\mathrm{osc}_i)$.

Proposition 5 (Privacy of one pseudo-posterior draw)**.** *With the DP weights, so that $|\alpha_i \tilde{\ell}_i(\Theta)| \le B$ on Θ_B, one exact draw $\Theta \sim p_\alpha(\cdot \mid D)$ is $2B$-DP for adding or removing a person, and $4B$-DP for replacing one. If the draw comes from a Markov chain within total-variation distance γ of p_α, the release is $(\varepsilon, \gamma(1 + e^\varepsilon))$-DP. Synthetic data generated from Θ alone (Algorithm 5) inherits the guarantee by post-processing.*

Proof. Since α_i depends only on D_i, adding person i changes the log-density by $\alpha_i \tilde{\ell}_i(\Theta) - \log \mathbb{E}_{p_\alpha(\cdot \mid D)} e^{\alpha_i \tilde{\ell}_i}$, and each term lies in $[-B, B]$. Replacement changes two such terms, giving $4B$. This is the exponential mechanism with a bounded utility [20, 5]. For the MCMC case, $P_D(S) \le P_D^*(S) + \gamma \le e^\varepsilon P_{D'}^*(S) + \gamma \le e^\varepsilon P_{D'}(S) + \gamma(1 + e^\varepsilon)$. □

Algorithm 6 Person-level weighted pseudo-posterior

Input: person map $i \mapsto D_i$; $r_{\max}$, value box, Θ_B, B (DP) or risk scores r_i (dial)
Output: weights α; weighted tuple statistics

1: keep at most $r_{\max}$ random rows per person per table; clip values ▷ SQL window `ROW_NUMBER`
2: compute osc_i over Θ_B (DP) or r_i (dial); set α_i
3: multiply person i's contributions to c, s, q and to the attribute statistics by α_i
4: run Algorithms 3–5 on the weighted statistics; release only Θ and synthetic rows, with new keys

Algorithm 7 RON-Gauss-style release of a high-dimensional block

Input: block X (person-bounded); target d; privacy (ε, δ) split over two releases
Output: DP mean and covariance in $\mathbb{R}^d$; synthetic block

1: $x_i \leftarrow x_i / \max(1, \|x_i\|_2)$ ▷ $O(\mathrm{nnz}(X))$
2: $W \leftarrow$ rows of $\mathrm{QR}(G)^\top$, $G \in \mathbb{R}^{p \times d}$ Gaussian ▷ $O(pd^2)$; data-independent, may be public
3: $y_i \leftarrow W x_i$ ▷ $O(\mathrm{nnz}(X)\, d)$
4: $\tilde{\mu} \leftarrow \frac{1}{n}(\sum y_i + \mathcal{N}(0, \sigma_1^2 I_d))$; $\tilde{\Sigma} \leftarrow \frac{1}{n}(\sum y_i y_i^\top + E) - \tilde{\mu}\tilde{\mu}^\top$, E symmetric with $\mathcal{N}(0, \sigma_2^2)$ entries
5: $\sigma_{1,2} = r_{\max}\sqrt{2\ln(1.25/\delta')}/\varepsilon'$ per release (Gaussian mechanism, $\varepsilon' < 1$), $\varepsilon = 2\varepsilon'$, $\delta = 2\delta'$
6: $\tilde{\Sigma} \leftarrow$ eigenvalue-clipped to PSD ▷ $O(d^3)$
7: synthesis: $y^* \sim \mathcal{N}(\tilde{\mu}, \tilde{\Sigma})$; $x^* = W^\top y^*$, then decode (threshold / arg-max for binary, sparse recovery for sparse blocks) ▷ $O(pd)$ per row

Complexity. Contribution bounding costs $O(N \log N)$ time with a sort, or $O(N)$ expected time with hashing. Weights cost $O(\sum_i |D_i|\, K)$ when osc_i is computed from per-term bounds on Θ_B. Sampling cost is unchanged.

Remark 9 (Utility cost). In the Poisson block model, a person's log-likelihood contains the exposure term $-\sum_l \eta_{kl} m_l$, whose range over Θ_B grows with the partner-table size. The oscillation can therefore be large, and α_i small, for every person. This is the utility cost of DP that layer 1 listed as an open risk. It has to be measured on real data. Noisy sufficient statistics [2] and the mechanisms of Hu, Williams and Savitsky [11] are the alternatives to compare against.

8 Algorithm 7: random orthonormal projection for sensitive high-dimensional blocks

Some columns form a block $X \in \mathbb{R}^{n \times p}$ with large p, such as wide one-hot or binary sets, embeddings of text or images, or sparse feature vectors. For these we follow RON-Gauss [3]: project to $d \ll p$ dimensions with a random *orthonormal* map, then release a Gaussian model through the Gaussian mechanism.

Proposition 6 (Sensitivity). *If $W \in \mathbb{R}^{d \times p}$ has orthonormal rows and $\|x\|_2 \le 1$, then $\|Wx\|_2 \le 1$ and $\|Wxx^\top W^\top\|_F = \|Wx\|_2^2 \le 1$. Hence $\sum_i y_i$ and $\sum_i y_i y_i^\top$ each have ℓ_2 (Frobenius) sensitivity at most 1 per row, and at most $r_{\max}$ per person.*

Proof. $WW^\top = I_d$, so $W^\top W$ is an orthogonal projection, and $\|Wx\|^2 = x^\top W^\top W x \le \|x\|^2$. Also $\|vv^\top\|_F = \|v\|^2$. □

Complexity. Time $O(\mathrm{nnz}(X)\, d + nd^2 + pd^2 + d^3)$ for the fit and $O(M(d^2 + pd))$ for synthesis. Space $O(pd + d^2)$, with the rows streamed. A sparse sign matrix (Achlioptas [1]; CountSketch for $s = 1$) reduces projection to $O(\mathrm{nnz}(X)\, s)$ but is not orthonormal, so its sensitivity bound must then be $\|R\|_2$ instead of 1.

Algorithm	Time	Space
1 Discovery (one pass)	$O(Np\,c_{\text{sk}}/P + Pp\,s_{\text{sk}})$	$O(p\,s_{\text{sk}} + \sum_j V_j + np)$
2 Copula fit	$O(I(np^2 + Tp^3))$	$O(np + p^2)$
3 Relational Gibbs	build $O(N_r)$; per iter. $O(\sum_r a_r\,\text{nnz}_r + \sum_e n_e \prod K + \sum_e n_e p_e^2)$	$O(\sum_r \text{nnz}_r + \sum_e n_e K_e)$
4 Merge	$O(K^2(L+p))$ per type and pass	$O(KL)$
5 Synthesis	$O(\sum_e n_e^*(K_e + c_f) + \sum_r a_r \mathbb{E}N_r^*)$	$O(\sum_e n_e^*)$, facts streamed
6 Privacy weights	$O(N \log N + \sum_i \lvert D_i \rvert K)$	$O(\#\text{persons})$
7 RP release	$O(\text{nnz}(X)d + nd^2 + pd^2 + d^3)$	$O(pd + d^2)$

Table 2: Complexity summary. $s_{\text{sk}} = k + \epsilon^{-1} \ln \delta^{-1} + m$ is the sketch size; $c_{\text{sk}} = \log k + \ln \delta^{-1}$ the per-update cost.

Distance preservation (compression and evaluation). For n points and $\epsilon \in (0, 1)$, a sparse $\{\pm\sqrt{3/d}, 0\}$ projection with

$$d \;\geq\; \frac{(4 + 2\beta) \ln n}{\epsilon^2/2 - \epsilon^3/3}$$

preserves every pairwise squared distance within $1 \pm \epsilon$ with probability at least $1 - n^{-\beta}$ [1]. Reconstruction $W^\top y$ is exact only for the part of x in the row space of W. It is therefore appropriate when the block is close to a d-dimensional subspace, or when it is s-sparse and $d = O(s \log(p/s))$ measurements allow compressed-sensing recovery.

9 Outer loop, evaluation, and end-to-end cost

Tuning (item 8). Hyper-parameters ξ (caps K_e, γ, η, B or risk weights, d) are tuned by Bayesian optimisation of $J(\xi) = U(\xi) - \lambda R(\xi)$, where U is utility and R is disclosure risk. With a Gaussian-process surrogate, T evaluations cost $O(T^3)$ on top of T runs of the pipeline.

Evaluation (item 10). The sliced-Wasserstein distance with L_s random directions costs $O(L_s\, n(p + \log n))$. Distance to closest record costs $O(n_{\text{syn}} n\, p)$ by brute force, or less with approximate nearest neighbours. The SyntheRela relational metrics [13] and Anonymeter [8] are used as published.

End to end. One database pass and one aggregate per relation, $O(\sum_T N_T p_T/P + \sum_r N_r)$, followed by I iterations at $O(\sum_r \text{nnz}_r + \sum_e n_e K_e L_e + np^2)$ and generation linear in the output. The only term that grows with the number of fact rows is paid once, inside the source database.

10 Validation

Numerical checks. Every checkable claim was tested by `validate_algorithms.py` (output in `validate_output.txt`). All 14 checks pass; Table 3 lists them.

Errors and gaps found and corrected.

1. **Rank-likelihood cost.** The layer-1 audit (§2.1) gives Hoff's sampler as $O(N \cdot p)$ per iteration. That counts only the latent draws. Each draw needs an $O(p)$ conditional mean, so a sweep is $\Theta(Np^2 + p^3)$ (T9). The two-stage conclusion stands and is strengthened.

2. **Sketch anchors for discrete columns.** The first run of T8 failed with empty latent intervals: KLL ranks of rare adjacent count levels coincided. Discrete columns now use exact level counts (Remark 3).

Check	Claim	Result
T1a, T1b	Prop. 2, (3) and (4)	match quadrature / direct MVN density to 10^{-9}
T2	Prop. 1 equals raw-fact likelihood incl. zero cells	max difference 10^{-13}
T3	Algs. 3+4 recover segment counts from caps $8, 6$	K=4, L=3 (true $4, 3$); accuracy 0.958 / 1.000
T4	per-iteration cost bounded by nnz, not $n_1 n_2$	nnz $\times 2.97 \Rightarrow$ time $\times 1.6$–1.9; 4×10^7 cells untouched
T5	Prop. 3	dispersion 0.997, cross-pair corr. 0.02
T6	Prop. 4	mean rows/customer 13.71 kept; 27.43 without rule
T7	C-vine construction gives LKJ marginals	KS $p = 0.09$, 0.69
T8	Alg. 2 on mixed data (gamma, ordinal, binary, count)	max $\lvert\hat{C} - C\rvert = 0.043$, $n = 3000$
T9	latent step is $O(np^2)$	log–log slope in p: 1.84
T10	merged KLL meets rank-error bound	$0.0027 \leq 0.0133$
T11	Prop. 5	max log-ratio $1.986 \leq 2B$ (add/remove), $2.42 \leq 4B$
T12	Prop. 6	bound 1.000 attained, never exceeded
T13	JL bound for sparse projection	ratios in $[0.83, 1.20]$ for $\epsilon = 0.3$

Table 3: Numerical validation (seeded; runtime about 30 s).

3. **JL constant.** A draft used $d = 8 \ln n/\epsilon^2$, which is below Achlioptas's bound at $\epsilon = 0.3$, and T13 failed. The stated bound is now the published one.

4. **Scaling.** "Scale = number of parent rows" (audit §3.1) silently changes per-entity cardinalities when a partner table is also resized. Proposition 4 gives the correction. Experiment 3 was unaffected because it kept the product count fixed.

5. **LKJ prior vs. Hoff's sampler.** Hoff's parameter-expanded inverse-Wishart update is not an LKJ prior, and it is invalid once anchors fix the latent scale. Algorithm 2 therefore uses Metropolis on canonical partial correlations with the exact LKJ prior (T7, T8). Feldman and Kowal use a factor model instead, which is a valid alternative.

6. **Stratified samples** must be self-weighting, or the copula likelihood must be survey-weighted.

7. **DP scope.** Proposition 5 holds for exact draws only; MCMC gives $(\varepsilon, \gamma(1 + e^{\varepsilon}))$. Its utility cost in the Poisson block model may be high.

What is not established. All tests use data generated from the model being tested. Self-referencing trees and temporal facts are specified but not run. Mixing at 10^8 fact rows, split–merge MCMC (as opposed to greedy merging) and the DP utility cost still need measuring in layer 2.

References

[1] D. Achlioptas. Database-friendly random projections: Johnson–Lindenstrauss with binary coins. *J. Comput. Syst. Sci.* 66(4):671–687, 2003.

[2] G. Bernstein, D. Sheldon. Differentially private Bayesian inference for exponential families. *NeurIPS*, 2018.

[3] T. Chanyaswad, C. Liu, P. Mittal. RON-Gauss: enhancing utility in non-interactive private data release. *PoPETs*, 2019.

[4] G. Cormode, S. Muthukrishnan. An improved data stream summary: the count-min sketch and its applications. *J. Algorithms* 55(1), 2005.

[5] C. Dimitrakakis, B. Nelson, Z. Zhang, A. Mitrokotsa, B. Rubinstein. Differential privacy for Bayesian inference through posterior sampling. *JMLR* 18, 2017.

[6] J. Feldman, D. R. Kowal. Using auxiliary marginal quantiles for Gaussian copula models with nonignorable missing data. *Bayesian Analysis*, 2025.

[7] P. Flajolet, É. Fusy, O. Gandouet, F. Meunier. HyperLogLog: the analysis of a near-optimal cardinality estimation algorithm. *AofA*, 2007.

[8] M. Giomi et al. A unified framework for quantifying privacy risk in synthetic data (Anonymeter). *PoPETs*, 2023.

[9] P. D. Hoff. Extending the rank likelihood for semiparametric copula estimation. *Ann. Appl. Stat.* 1(1), 2007.

[10] J. Hu, T. D. Savitsky, M. R. Williams. Risk-efficient Bayesian pseudo-posterior data synthesis for privacy protection. *J. Survey Stat. Methodol.*, 2022.

[11] J. Hu, M. R. Williams, T. D. Savitsky. Mechanisms for global differential privacy under Bayesian data synthesis. *Statistica Sinica* 35, 2025.

[12] S. Jain, R. M. Neal. A split-merge Markov chain Monte Carlo procedure for the Dirichlet process mixture model. *J. Comput. Graph. Stat.* 13(1), 2004.

[13] M. Jurkovic, V. Hudovernik, E. Štrumbelj. SyntheRela: a benchmark for synthetic relational database generation. ICLR workshop, 2025.

[14] Z. Karnin, K. Lang, E. Liberty. Optimal quantile approximation in streams. *FOCS*, 2016.

[15] C. Kemp, J. Tenenbaum, T. Griffiths, T. Yamada, N. Ueda. Learning systems of concepts with an infinite relational model. *AAAI*, 2006.

[16] D. Lewandowski, D. Kurowicka, H. Joe. Generating random correlation matrices based on vines and extended onion method. *J. Multivariate Anal.* 100(9), 2009.

[17] J. Rousseau, K. Mengersen. Asymptotic behaviour of the posterior distribution in overfitted mixture models. *JRSS-B* 73(5), 2011.

[18] F. Saad, V. Mansinghka. Hierarchical infinite relational model. *UAI*, 2021.

[19] T. D. Savitsky, D. Toth. Bayesian estimation under informative sampling. *Electron. J. Stat.* 10(1), 2016.

[20] Y.-X. Wang, S. Fienberg, A. Smola. Privacy for free: posterior sampling and stochastic gradient Monte Carlo. *ICML*, 2015.

[21] R. J. Wilson et al. Differentially private SQL with bounded user contribution. *PoPETs*, 2020.

[22] K. Zhang, N. Patki, K. Veeramachaneni. Sequential models in the Synthetic Data Vault. arXiv:2207.14406, 2022.